

Ulugbek Abdullaev

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Education

Purdue University

Master of Science in Electrical and Computer Engineering

- Relevant Graduate Coursework: Connected and Automated Vehicles, Multi-Agent Autonomy and Control, Lumped Linear Systems, Dynamical Systems

West Lafayette, IN

Aug 2024 – Aug 2026

Tashkent State Transport University

Bachelor of Science in Automation and Control

Tashkent, Uzbekistan

Sep 2019 – Jun 2023

Professional Experience

Controls Engineering Intern

Tesla, Inc.

Brooklyn Park, MN

Aug 2025 – Jan 2026

- Developed Siemens S7-1500 PLC ladder logic in TIA Portal to integrate 50+ hardware devices, including actuator clamps, KEYENCE barcode readers, SICK proximity sensors, and Festo valve manifolds
- Executed the hardware configuration for devices in Siemens TIA Portal, assigning static IP addresses and validating robust PROFINET communication links across the entire network
- Designed 20+ operator-centric HMI screens in Siemens SIMATIC WinCC with custom faceplates for 3 assembly lines, delivering real-time process visualization and enabling precise operator control over production
- Developed sequential control logic using Siemens S7-GRAPH (SFC) for the takeout and transfer nests, enabling manual override and controlled recovery during process upsets to minimize production downtime
- Programmed safety PLC ladder logic for emergency stops, gate switches, and safety control valves, configuring PROFIsafe addresses on Festo valve manifolds to ensure compliance with industrial safety standards

Controls Engineering Intern

Schreiber Foods

Monett, MO

Jun 2025 – Aug 2025

- Engineered ladder logic in Rockwell Studio 5000 to transition Hayssen wrappers from manual to full automation, resolving feed imbalances and reducing cheese salvage for an estimated \$500k/year savings
- Integrated a KEYENCE Level Sensor into the vacuum hopper and engineered ladder logic in Rockwell Studio 5000 to automate valve control for liquid cheese distribution to casting belts
- Implemented SATRON VOA optical sensor to precisely regulate cheese flow from cookers, ensuring accurate diversion decisions and reducing waste by 25% caused by manual draining
- Updated electrical schematics in AutoCAD to reflect new equipment installations, ensuring accurate, up-to-date documentation for controls and maintenance teams

Controls Engineering Intern

Uzbekistan Railways

Tashkent, Uzbekistan

Feb 2023 – May 2023

- Designed relay-based control circuits using Microsoft Visio to enhance train movement safety and efficiency
- Developed PLC ladder logic to process axle counter inputs and programmed route-setting algorithms using Structured Text, reducing signal processing latency by 25% and eliminating manual interventions
- Integrated inductive proximity sensors and Siemens ACM 250 axle counter into control systems, enabling real-time data collection and improving system responsiveness by 15%

Controls Engineering Intern

Uzbekistan Railways

Bukhara, Uzbekistan

Jun 2022 – Aug 2022

- Developed sorting algorithms in collaboration with senior engineers to streamline the sorting processes
- Performed preventive maintenance on traffic lights and control relays, minimizing system downtime by 20%
- Contributed to redesigning the sorting station by applying optimization methods, resulting in a 35% cost reduction

Technical Skills

PLC/Automation: Siemens S7-1200 / S7-1500 (TIA Portal), Allen-Bradley ControlLogix / CompactLogix (Studio 5000), SCADA, PLC Ladder Logic, Structured Text, SFC (S7-GRAPH), HMI Development (WinCC), Safety PLC Programming, IO Configuration

Industrial Networking: PROFINET, PROFIsafe, EtherNet/IP, Troubleshooting, Fault Diagnostics

Programming Languages/Tools: Python, MATLAB/Simulink, AutoCAD, Bluebeam, Microsoft Office (Word, Excel, PowerPoint, Visio)

Publications

“Methods for increasing the capacity of sorting processes on railway sorting hills”, Z. Toshboev, U. Abdullaev, Universum, Dec 2022